BUSINESS ANALYSIS CASE STUDY · SHORT-SEA & DOOR-TO-DOOR LOGISTICS

Shipment exceptions, handled before the customer calls

How a fictional operator could stop its customers from learning about delays at the gate — from a framed problem and a business case, through workshops, rules, backlog and design, to acceptance testing and a controlled go-live.

Independent portfolio case. Tidewell Logistics is fictional. People, workshops, sprints and acceptance tests are simulated; figures are illustrative assumptions. Results are labelled proposed (business case), simulated (fictional delivery) or measured (automated tests actually run in the repository). Nothing here was deployed in a real organisation. Repository: github.com/phlppgdfry/shipment-exception-management-business-analysis

1. Executive summary

Since its new transport management system (TMS) went live, Tidewell's customers can see statuses in a portal — but not whether their **agreed delivery window** is at risk. Only 22% of late deliveries were announced before the window started; "where is my load?" was 38% of all customer contacts. Three managers asked for three solutions: a tracking page, e-mails to key accounts, and "nothing that adds work".

The analysis showed that the gap was not a missing screen but **missing rules**: what counts as late, how serious it is, who acts and who tells the customer. The recommended option is a small exception component next to the TMS: rules decide severity, owner and response time; planners work from one list; customers get one clear message. Because its value rests on one uncertain number, the sponsor funded a **six-week pilot with a stage gate** rather than the full build.

22% → ≥90%
major/critical delays told before the window — proposed

–35%
WISMO contacts vs. a control group — proposed

13.5 mo
payback; pessimistic case ±31 months — proposed

100%
automated checks passing in the repository at the time of writing — see the test-results page — measured

2. Business challenge

What hurts, for whom

WHO	PAIN TODAY
Customer	Plans docks around the window; hears of delays at the gate — "rather bad news at 10 than no news at 4"
Customer service	±1,250 WISMO contacts a week; a "switchboard" between customer and planner
Road planner	Notices delays by chance; copies orders into an Excel sheet; interrupted by calls
Account manager	Hears about problems from the customer
Finance	±55 missed slots a week at ±€140; credit notes without a reason

Root cause (5 whys)

The customer calls → nobody told them → customer service did not know either → nobody owns "telling the customer" → every TMS change looks the same, so people filter by gut feeling → **there is no shared definition of late, no severity and no owner**.

What would change our mind

If most WISMO calls concerned on-time orders, a status page would be the fix. A sample showed ±70% were about late or at-risk orders.

3. Discovery and analysis approach

Eight interviews, a shadowed morning shift and a two-week contact-reason tagging exercise produced a baseline and five contradictions between teams. Instead of writing one person's view into a process diagram, the analyst ran a **three-hour big-picture event storming** with planners, customer service, an account manager, the customs desk, the Product Owner and a developer. Disagreements became six hotspots, each with a decision owner and a date.

Four options, scored on agreed weights

OPTION	SCORE	VERDICT
O1 Add a customer-service agent	2.45	treats the symptom
O2 Standard TMS e-mails	3.10	±9 changes per order, ±0.3 relevant: noise
O3 Exception management	**3.95**	**recommended, as a pilot**
O4 External visibility platform	2.85	later, for consignees

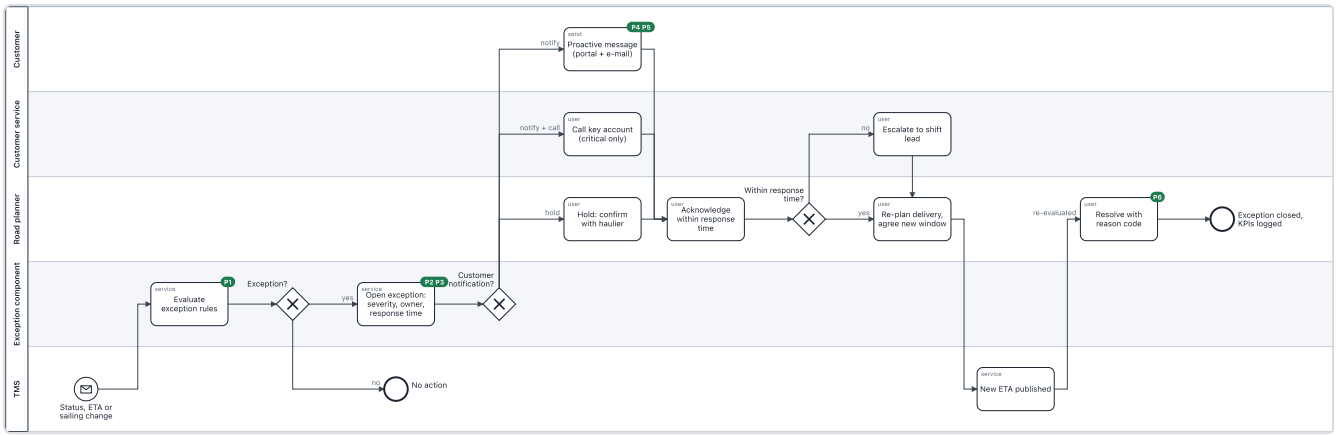
The decision that unlocked the design

"Tell key accounts everything" and "no message storm" looked like opposites. A concrete story from each side revealed the needs behind them — *never surprise a key account with something serious* and *no message that doesn't change the customer's day*. **Severity** served both. The Product Owner decided (D-02); the analyst prepared options and consequences.

Artefacts: problem statement, capability map and SWOT, business case with sensitivity, KPI tree, stakeholder map and RACI, interview notes, event-storming board, decision log, assumption/risk/dependency registers, research register.

4. Proposed process improvement

The AS-IS (BPMN 2.0) showed that the information exists — the sailing system knows a delay within minutes — but is lost between people. The TO-BE adds **rules and ownership**, not a tracking screen.



TO-BE, release 1 — every TMS change is evaluated by the rules; exceptions get an owner and a response time; the customer is told proactively. Valid BPMN 2.0 XML in the repository.

RULE	DECISION
ER-01	An exception exists only when expected delivery is after the **end** of the agreed window (or a cancellation, customs hold or missing status)
ER-02/03/04	1–119 min minor · 120–359 major · ≥360 or next day critical · reefer and time-critical one level up
ER-07	Missing status is held until a planner confirms with the haulier — no false alarms
ER-09	Respond within 30 min (critical) / 2 h (major), otherwise escalate to the shift lead
ER-10/11	Major and critical are notified; critical for a key account also creates a call task
ER-12/13	At most one message per order every 2 h unless worse; "back on schedule" only if told before

The thirteen rules live in one table and one code module. Their thirty Given/When/Then examples **run as tests** — so when the PO changes a threshold, the test run shows which decisions change meaning.

5. Requirements and product delivery

Five business requirements, seventeen functional and nine non-functional requirements, each with an ID and a verification method, trace from pain point to KPI:

> *P5 "customer finds out at the slot" → BR-1 → FR-06, FR-10 → ER-10, ER-12 → US-05, US-07 → AC-01.6, AC-07.1, BAT-07/08 → K1, K3, K8*

With the Product Owner, the backlog was ordered by **WSJF**: when everything was "Must", it surfaced small, high-value items (making the delivery window mandatory scored highest — 28% of orders had none). Dependencies then set the build order. Release 1 = 36 points, two sprints, five pilot customers on two routes, behind a feature toggle.

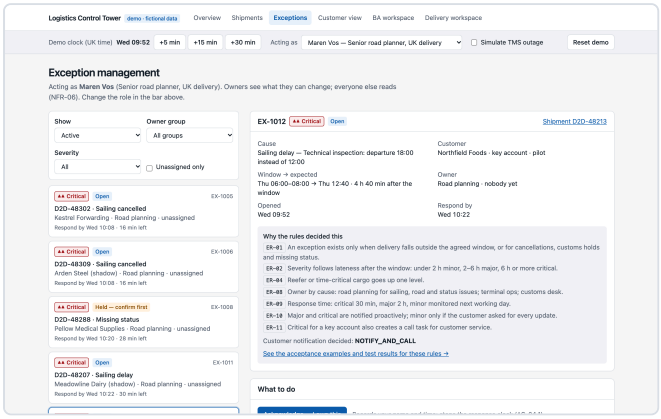
Who decided what

DECISION	OWNER
What counts as late (D-01)	PO
Who is told what (D-02)	PO
Build small vs. wait for supplier (D-03)	PO + architect
Missing status: hold first (D-04)	Road planning lead
No consignee messages yet (D-05)	DPO + commercial
Pilot and stage gate (D-07)	Sponsor + PO

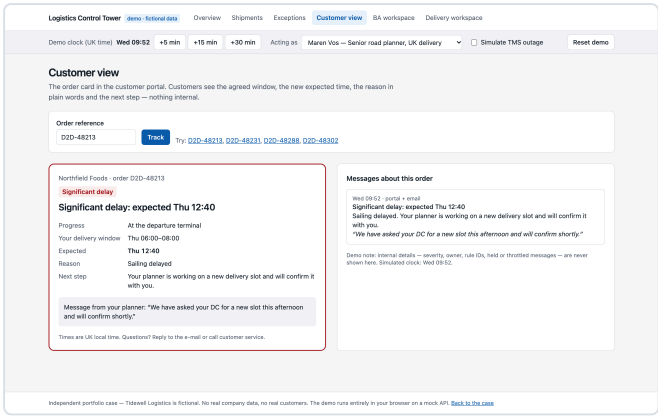
The analyst is responsible for much and accountable for little: scope, priority and go-live stay with the PO.

6. UX and technical collaboration

Personas and journey maps located the *moments that matter*: the planner must trust the severity, and the customer's message must be clear and in UK local time. The planner workbench and customer messages were designed in Figma; a working **control tower** demo then implemented the release on the same rule module, with a guided ten-step scenario.



Workbench: most urgent first; every exception explains which rules decided it.



Customer card: built only from messages actually sent — nothing internal.

For developers and architects, the analysis includes a conceptual architecture (proposed vs. implemented), an OpenAPI 3.1 contract, idempotency and retry rules, a data model, state machines and a role matrix. **Building the demo was itself a review:** it exposed five open questions the specification had missed — such as an exception that recovers before anyone acknowledged it — and a privacy leak in the first customer API, fixed with a public message projection and a test.

7. Testing and rollout readiness

Acceptance simulated

Business Acceptance Testing with key users — deliberately including the sceptical senior planner — found six issues. Three of four majors came from people reading real messages in their own context; the team's tests had passed. One came from the analyst's own refinement answer (throttling per exception instead of per order) and became a retrospective action. Sign-off with two known issues, each with a workaround, owner and date.

Go-live simulated

Thirteen go/no-go criteria with an owner and evidence each; two days in shadow mode; customers switched on one by one on a Monday; rollback rehearsed in four minutes; hypercare thresholds; a support article that answers "rule or bug?".

Automated evidence measured

SUITE	WHAT IT PROVES
Rule examples (node:test)	30 decision examples + coverage and monotonic-threshold checks
Engine & contract (Vitest)	lifecycle, permissions, idempotency, API responses vs. OpenAPI
Browser (Playwright + axe)	the full scenario, workbench behaviours, all routes on desktop and mobile, WCAG 2.2 AA automated checks
Repository gate	every ID defined and referenced correctly; traceability; BPMN validity

Exact counts, run date and commit: docs/09-acceptance/test-results.md. CI runs them on every push.

8. Lessons learned and next steps

Lessons

- **Frame before solving.** Three requested solutions hid one missing definition.
- **Buy the information first.** A pilot with a control group tests the riskiest assumption before the full investment.
- **Make rules executable.** Examples that run as tests keep business, code and specification aligned.
- **Passing stories ≠ met need.** Real readers in their own context find what tests cannot.
- **Building is reviewing.** Implementing the lifecycle found gaps no document review had.

Next steps (in the case)

- Stage gate after six pilot weeks: fund release 2 if WISMO falls ≥20% vs. control.
- Release 2: EDI/API status messages, customs holds with customer action, notification preferences, consignee opening hours.
- Resolve open questions OQ-01 – OQ-05 with their owners.

Limitations

Fictional organisation and simulated delivery; no real TMS, e-mail or load testing; architecture and privacy controls not reviewed by real experts. The case shows how the analyst works and decides — it is not a substitute for team experience.

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